Caron 2020 Acute Lymphoblastic Leukemia — analysis summary

RUN 3 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) fuses the three genetically distinct subtypes: same-sample neighbour fraction 1.00→0.26, iLISI 1.0→4.2/8 (sample) and 2.5/3 (subtype), one connected kNN component at n_pcs=20/30/40. B-ALL (ETV6-RUNX1, HHD) and T-ALL intermix on one continuous manifold; the only within-manifold structure is a cycling↔quiescent axis (both poles carry all 3 subtypes), not a lineage split.

Gate 2 — module geography → GO (non-branching)

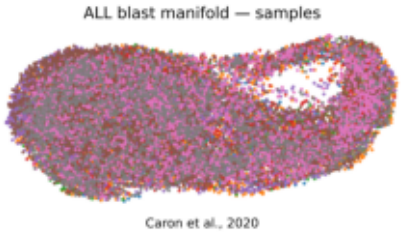
Signatures collapse onto a single (Gast/pSom)→Crest axis: smoothed pSom↔Crest r=−0.74, Gast↔pSom r=−0.46 (organized cross-repression), Gast↔Crest r=−0.20. Gast and pSom sit at OPPOSITE poles — no co-expression hub, no two competing arms. Crest is a terminal gradient (joint pSom&Crest enrichment 0.28 < 1). Along pseudotime Gast falls (ρ=−0.73), Crest rises (decile pt 0.68). Curve fold-free (2 tips / 0 forks). Crest does NOT enrich a Gast arm — no flag.

Gate 3 — hub cross-repression → NA

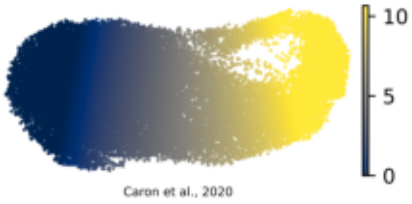
Non-branching topology — no bifurcation and no Gast/pSom hub, so the branching-only cross-repression test does not apply. Correlation & modules notebooks are correctly not produced.

STEP FIGURES

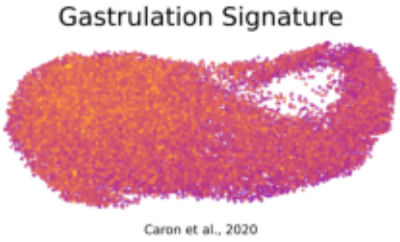
Manifold — samples



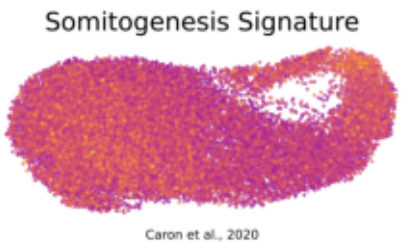
Pseudotime



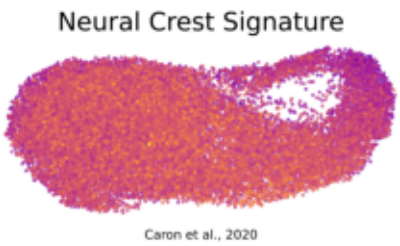
Gastrulation



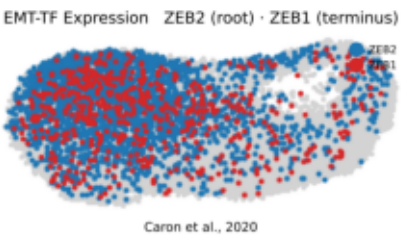
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric ALL (Caron et al., 2020), 10x — 25,784 malignant lymphoblasts over 8 samples spanning B-ALL (ETV6-RUNX1), B-ALL (HHD) and T-ALL.

Verdict.
Pattern reproduces as the NON-BRANCHING case — one gastrulation/somitogenesis → neural-crest developmental axis with organized Gast↔pSom cross-repression; no hub, no bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means (raw) Gast −0.017 / pSom +0.000 / Crest −0.010 — Linear. EMT-TFs ZEB2 (root, 40.6%) → ZEB1 (terminus, 8.9%).

Parameter choices

Min-cell threshold	100 floor — counts fall smoothly (no dominant cliff); all 8 samples kept	**Integration**	Harmony θ=10 (mixing plateaus θ=10–15, θ=20 regresses); n_pcs=30, n_neighbors=30, spread=1.3
Islands removed	4 cells / 3 detached specks (UMAP radius graph, largest component)	**Geometry**	Non-branching scFates curve (ElPiGraph), Nodes=20 λ=0.3 μ=100; 2 tips / 0 forks, fold-free
Root node	curve tip 0 = Gast/pSom-high, Crest-low (origin; Gast→Crest along pseudotime)	**Module means**	Gast −0.017 / pSom +0.000 / Crest −0.010 — Linear
EMT-TFs	ZEB2 (root, 40.6%) → ZEB1 (terminus, 8.9%); SNAI2/TWIST1/FOXC2 <0.1% gated out		